

COMPUTER NETWORK

Course Code: BCA-503

Exam-Friendly Study Notes | Unit I and Unit II

Programme	BCA (Bachelor of Computer Applications)
University	Chaudhary Charan Singh University (CCSU), Meerut
Unit I	Basic Concepts, OSI and TCP/IP Models, DTE-DCE Interface, Modems, Cable Modems
Unit II	Transmission Media, Impairments, Throughput, Propagation, Wavelength, Shannon Capacity

Written in simple English for quick understanding, answer writing and last-day revision.

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UNIT I: Basic Concepts, OSI/TCP-IP Models, Interfaces and Modems

1.1 Components of Data Communication

Definition: Data communication is the exchange of data between two devices through a transmission medium (wire or air).

Easy Meaning: Sending information from one device to another using a cable or a wireless path.

Component	Meaning	Example
Message	The information to be sent (text, number, image, audio, video)	E-mail, photo
Sender	Device that sends the message	Computer, mobile, camera
Receiver	Device that receives the message	Computer, printer, TV
Transmission medium	Physical path on which the message travels	Twisted pair, fibre, radio waves
Protocol	Set of rules that both devices follow. Without it, devices are connected but cannot understand each other (like two people speaking different languages)	TCP/IP, HTTP

Key Points (qualities of good data communication):

- **Delivery** - data reaches the correct destination only.
- **Accuracy** - data is delivered without change (no errors).
- **Timeliness** - data is delivered in time (very important for live audio/video).
- **Jitter** - uneven delay in arrival of packets; it should be low.

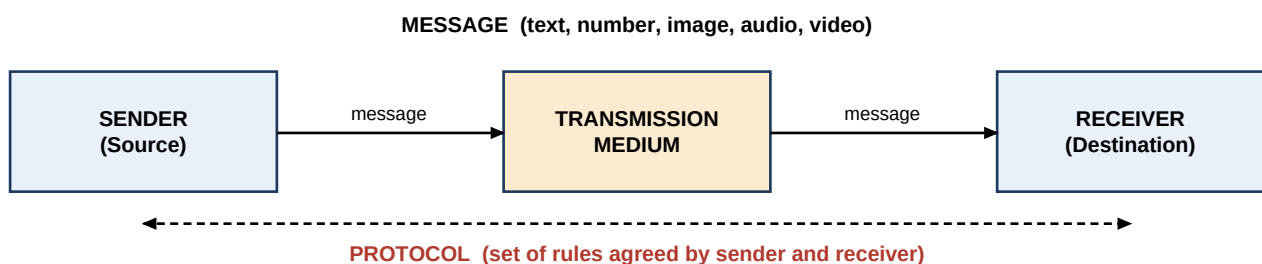


Fig 1.1: Five components of data communication

How to Write in Exam: 1) Definition 2) List five components with a table 3) Diagram 4) Four qualities (delivery, accuracy, timeliness, jitter).

Keywords: message, sender, receiver, medium, protocol, delivery, accuracy, timeliness, jitter

1.2 Distributed Processing

Definition: Distributed processing is a method in which a task is divided among many computers instead of one central machine doing everything.

Simple Explanation: Each computer does a part of the work. Most networks today use it: a big database or job is shared among many machines.

Example: World Wide Web: web servers, database servers and your browser share the work of showing one web page.

- **Security/encapsulation** - a user can access only the part of a system he is allowed to.
- **Distributed databases** - no single machine holds all the data.
- **Faster problem solving** - many computers work on a problem together.
- **Collaborative processing** - many computers work together (e.g. multiplayer games).

Advantages: Better speed, less load on one machine, reliability (if one fails, others work), easy expansion.

Disadvantage: Complex to manage and needs a good network.

1.3 Standards and Organizations

Definition: A standard is an agreed set of rules/specifications so that devices of different makers can work together (interoperability).

Type of standard	Meaning	Example
De facto ("by fact")	Not officially approved, but widely used in practice	TCP/IP (started in practice), Windows
De jure ("by law")	Officially approved by a recognized body	OSI model (ISO)

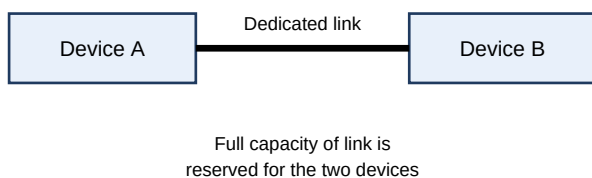
Organization	Full form	Role
ISO	International Organization for Standardization	Made the OSI model; global standards
ITU-T	International Telecommunication Union - Telecommunication Standardization Sector (earlier CCITT)	Standards for telephone and data communication (e.g. X.21, V.90)
ANSI	American National Standards Institute	Official US standards body
IEEE	Institute of Electrical and Electronics Engineers	LAN standards (802 series: Ethernet 802.3, Wi-Fi 802.11)
EIA	Electronic Industries Alliance	Interface standards such as RS-232 (EIA-232)
IETF	Internet Engineering Task Force	Internet protocols (documents called RFCs)
FCC	Federal Communications Commission (USA)	Regulatory agency for radio, TV, telephone

1.4 Line Configuration

Definition: Line configuration is the way two or more devices are connected to a link. A link is the communication path between devices.

Point	Point-to-Point	Multipoint (Multidrop)
Meaning	Dedicated link between exactly two devices	One link shared by three or more devices
Capacity	Full capacity used by the two devices	Shared: spatially (used together) or time-shared (turn by turn)
Cost	More cables, costlier	Less cable, cheaper
Example	Computer to printer; remote control to TV	Mainframe (host) with many terminals

(a) Point-to-Point



(b) Multipoint (Multidrop)

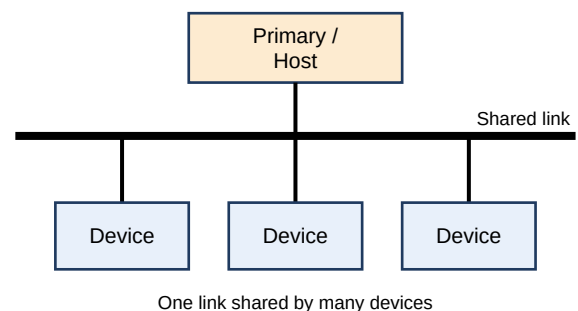
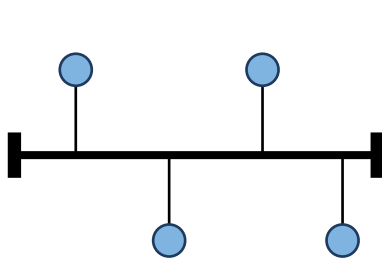


Fig 1.2: Line configurations

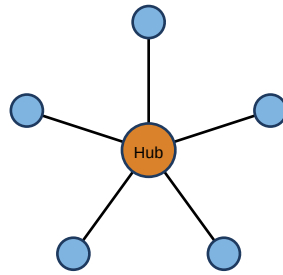
1.5 Topology

Definition: Topology is the geometric arrangement of links and nodes (devices) in a network.

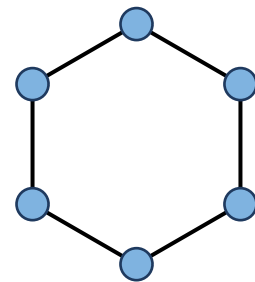
Simple Explanation: It shows how computers are connected to each other: as a line, star, circle, etc.



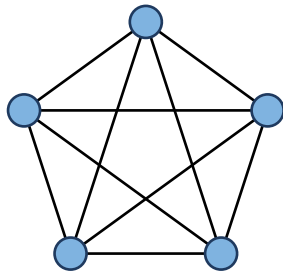
BUS (backbone + terminators)



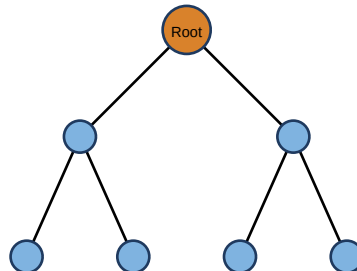
STAR



RING



MESH



TREE

HYBRID

Combination of two or more topologies

e.g. Star + Bus

Fig 1.3: Network topologies

Topology	How it works	Advantages	Disadvantages
Mesh	Every device has a dedicated point-to-point link to every other device	No traffic problem; robust (one link fails, others work); secure/private; easy fault finding	Too much cabling and many I/O ports; costly; hard to install
Star	Every device connects only to a central hub/switch; no direct device-to-device link	Cheap, easy to install and extend; failure of one link affects only that device	If hub fails, whole network fails; more cable than bus
Bus	One long backbone cable; devices join through drop lines and taps; terminators at both ends	Easy to install; least cable; cheap	Difficult to add devices; break in backbone stops all; signal reflection; limited length
Ring	Each device is joined to exactly two neighbours; signal moves in one direction (token)	Easy to install and reconfigure; easy fault isolation	Break in ring stops network (unless dual ring); one-way traffic
Tree	Hierarchy of stars; a central root node with branches	Scalable; easy expansion; branch-wise fault isolation	Root failure breaks network; more cabling; complex

Numerical: Mesh topology

Formula: Duplex links = $n(n-1)/2$ and ports per device = $n-1$, where n = number of devices.

Example: $n = 5$. Links = $5 \times 4 / 2 = 10$. Ports on each device = $5 - 1 = 4$.

Common mistakes: Forgetting to divide by 2; using n links per device instead of $n-1$.

How to Write in Exam: 1) Definition 2) Diagram of each topology 3) Working 4) Advantages and disadvantages (table).

Keywords: physical/logical arrangement, node, link, backbone, hub, terminator, dedicated link, $n(n-1)/2$

1.6 Transmission Mode

Definition: Transmission mode (direction of data flow) tells in which direction signals can travel between two connected devices.

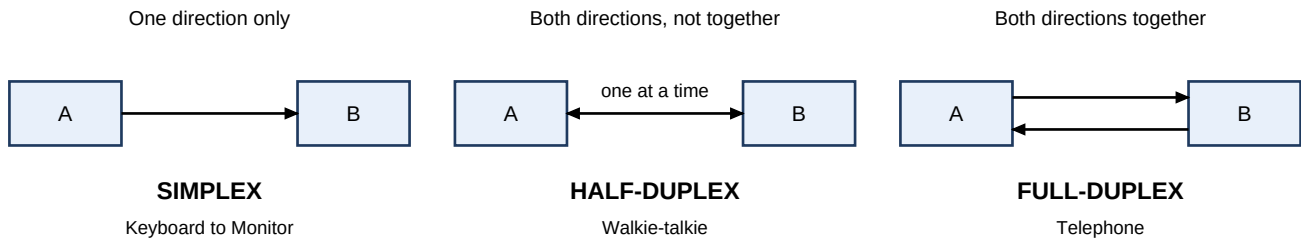


Fig 1.4: Transmission modes

Mode	Direction	Example	Note
Simplex	One-way only (sender always sends, receiver only receives)	Keyboard, monitor, radio/TV broadcast	Uses full capacity of channel in one direction
Half-duplex	Both ways, but one at a time	Walkie-talkie	Whole capacity used by one side at a time
Full-duplex	Both ways at the same time	Telephone	Capacity is shared (or two separate paths used)

1.7 Categories of Networks

Definition: A computer network is a set of devices (nodes) connected by links to share data and resources. Networks are classified by their size/area.

Point	LAN	MAN	WAN
Full form	Local Area Network	Metropolitan Area Network	Wide Area Network
Area covered	Room, building, campus (up to a few km)	A city (up to about 50-100 km)	Country, continent, world
Ownership	Private (one organization)	Private or public	Many organizations/ public carriers
Speed	High	Medium	Lower than LAN
Cost / error rate	Low cost, few errors	Medium	High cost, more errors
Example	College computer lab	Cable TV network of a city	The Internet, bank networks

Internetwork (internet): Two or more networks joined using routers form an internetwork. The Internet is the largest one.

How to Write in Exam: 1) Definition 2) Table LAN/MAN/WAN 3) Example of each.

Keywords: area, ownership, speed, cost, private, public

1.8 OSI Model

Definition: The OSI (Open Systems Interconnection) model, developed by ISO, is a 7-layer reference model that explains how data moves from one computer to another in a network.

Simple Explanation: Big job of communication is divided into 7 smaller jobs (layers). Each layer serves the layer above and uses the layer below.

Memory trick (layer 1 to 7): Please Do Not Throw Sausage Pizza Away.

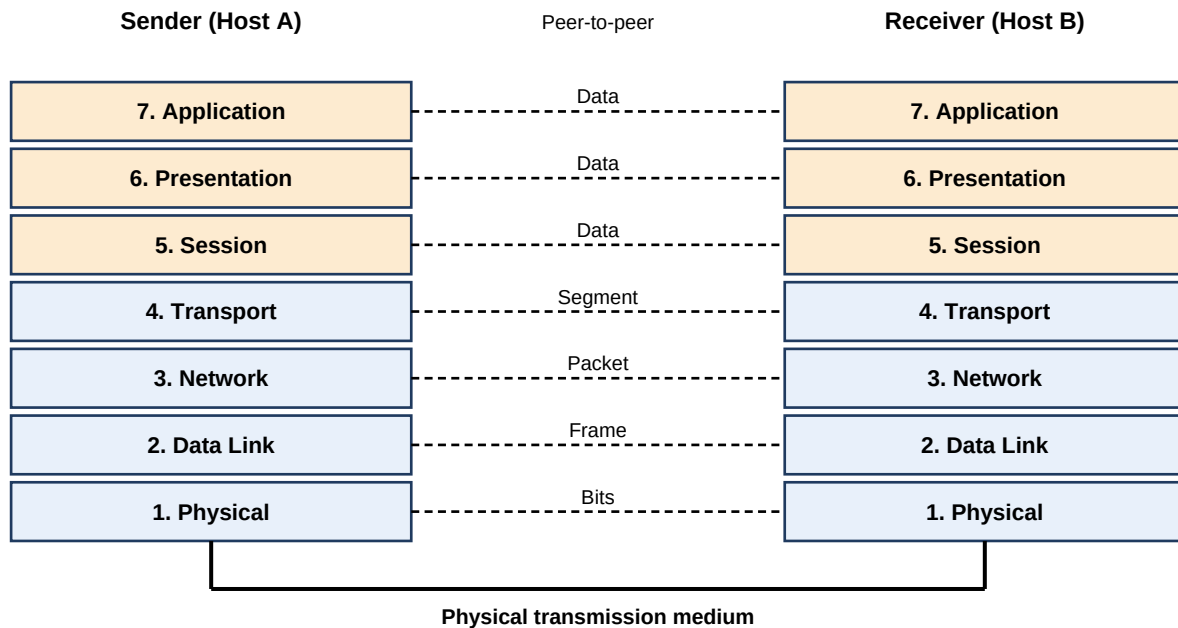


Fig 1.5: OSI model (orange = user-support layers 5-7, blue = layers 1-4)

- **Peer-to-peer process:** a layer on sender talks to the same layer on receiver (using that layer's protocol).
- **Encapsulation:** on the sender side each layer adds its header (layer 2 also adds a trailer); the receiver removes them layer by layer.
- Layers 1-3 are **network-support** layers; layers 5-7 are **user-support** layers; layer 4 links the two groups.

Layer	Functions	PDU / Examples
7 Application	Gives network services to the user: file transfer, e-mail, remote login, directory services, World Wide Web.	Data HTTP, FTP, SMTP, Telnet
6 Presentation	Translation (different codes such as ASCII/EBCDIC), encryption/decryption (security), compression (reduce size).	Data
5 Session	Dialog control (who sends when), synchronization (checkpoints), establishing, maintaining and ending a session.	Data
4 Transport	Process-to-process (end-to-end) delivery; port (service-point) addressing; segmentation and reassembly; connection control; flow control; error control.	Segment TCP, UDP
3 Network	Source-to-destination delivery across networks; logical (IP) addressing; routing; packetizing.	Packet Router
2 Data Link	Node-to-node delivery; framing; physical (MAC) addressing; flow control; error control; access control.	Frame Switch, Bridge, NIC
1 Physical	Sends raw bits over the medium: physical characteristics of interface and medium, representation of bits (encoding), data rate, synchronization of bits, line configuration, topology, transmission mode.	Bits Cables, Hub, Repeater

How to Write in Exam: 1) Definition 2) Diagram of 7 layers 3) Function of each layer (table) 4) Encapsulation/peer-to-peer note.

Keywords: ISO, seven layers, reference model, peer-to-peer, encapsulation, framing, routing, port address, dialog control, encryption, translation

1.9 TCP/IP Model

Definition: The TCP/IP model is the protocol suite used by the Internet. It was developed by DARPA (US Department of Defense) and has 4 layers.

Simple Explanation: It is a shorter, practical version of OSI. Protocols came first and the model was described later.

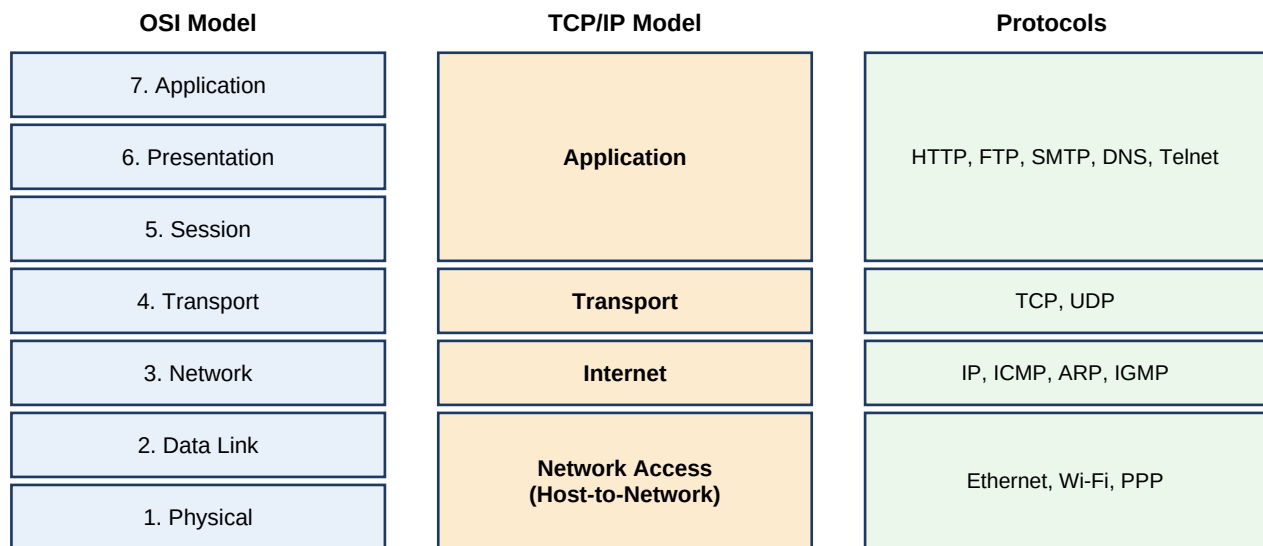


Fig 1.6: OSI and TCP/IP layers

Layer	Functions	Protocols
Application	Provides services to the user. It does the work of OSI application, presentation and session layers.	HTTP, FTP, SMTP, DNS, Telnet
Transport	End-to-end delivery, segmentation, flow and error control. TCP is reliable and connection-oriented; UDP is unreliable and connectionless.	TCP, UDP
Internet	Logical (IP) addressing, routing and fragmentation of packets across networks.	IP, ICMP, ARP, IGMP
Network Access (Host-to-Network)	Physical addressing, framing and sending bits over the physical medium.	Ethernet, Wi-Fi, PPP

Note: Some books show 5 layers (Network Access split into Physical + Data Link). Follow your textbook, but the 4-layer form above is the most common.

1.10 Comparison of OSI and TCP/IP Models

Point	OSI Model	TCP/IP Model
Full form / developer	Open Systems Interconnection; ISO	Transmission Control Protocol/Internet Protocol; DARPA
Layers	7	4 (or 5)
Approach	Model first, protocols later; protocol independent	Protocols first, model later; protocol dependent
Session and presentation	Separate layers	Part of application layer
Network layer service	Connection-oriented and connectionless	Connectionless only (IP)
Transport layer service	Connection-oriented only	Both (TCP and UDP)
Use	Mainly a reference/teaching model	Actually used on the Internet

Similarities: Both are layered; both have application, transport and network layers; both use packet-based communication.

How to Write in Exam: Write a comparison table with at least 6 points, then draw the layer-mapping diagram (Fig 1.6).
Keywords: reference model, protocol suite, connection-oriented, connectionless

1.11 DTE-DCE Interface

Definition: **DTE** (Data Terminal Equipment) is any device that generates or receives digital data, e.g. computer. **DCE** (Data Circuit-terminating Equipment) is the device that connects the DTE to the network and converts the signal, e.g. modem. The DTE-DCE interface is the physical connection between them, defined by a standard.

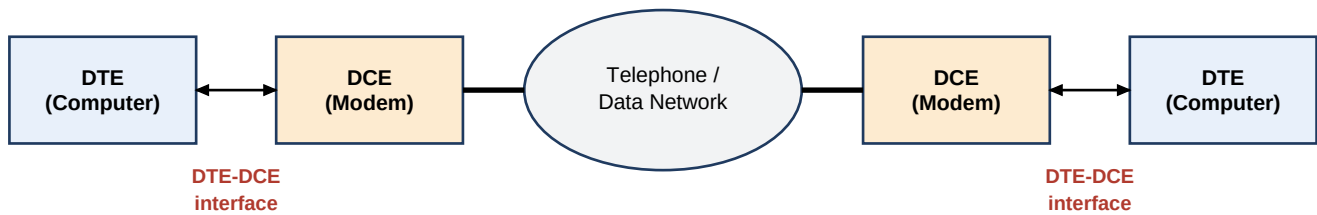


Fig 1.7: DTE-DCE interface

Characteristic	Meaning
Mechanical	Size, shape and number of pins of the connector
Electrical	Voltage levels, timing and data rate
Functional	Function of each pin/circuit (data, control, timing, ground)
Procedural	Sequence of events for sending data

Standard	Details
EIA-232 (RS-232)	25-pin (DB-25) or 9-pin (DB-9) connector. Binary 1 = -3 to -15 V, binary 0 = +3 to +15 V. Short distance (about 15 m) and low speed (about 20 kbps).
X.21 (ITU-T)	15-pin connector for digital networks; used for synchronous digital communication.

Main RS-232 (DB-25) pins: Pin 2 TxD (transmit data), 3 RxD (receive data), 4 RTS (request to send), 5 CTS (clear to send), 6 DSR (data set ready), 7 signal ground, 8 DCD (carrier detect), 20 DTR (data terminal ready).

1.12 Modems

Definition: A modem (MODulator + DEModulator) converts digital data of a computer into analog signals for a telephone line, and converts received analog signals back into digital data.

Simple Explanation: Computer speaks in 0s and 1s, but the telephone line carries only sound-like waves. A modem is the translator between the two.

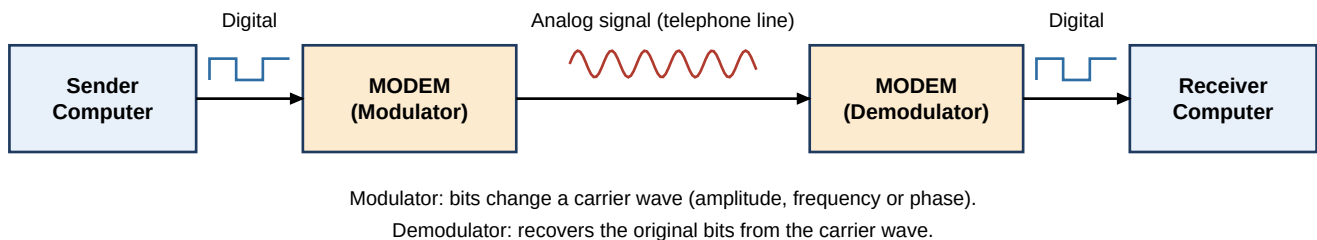


Fig 1.8: Working of a modem

Modulation	What changes in the carrier wave
ASK	Amplitude Shift Keying - amplitude
FSK	Frequency Shift Keying - frequency
PSK	Phase Shift Keying - phase
QAM	Quadrature Amplitude Modulation - amplitude and phase together (used in high-speed modems)

- **Types:** internal/external; dial-up (telephone), DSL, cable, wireless.
- **56K modem (V.90):** about 56 kbps download and 33.6 kbps upload on a telephone line.

Advantages: Uses existing telephone line, low cost.

Disadvantages: Slow; phone line is busy during use; affected by line noise.

Applications: Dial-up Internet, fax, remote login.

How to Write in Exam: 1) Definition (modulator + demodulator) 2) Diagram 3) Working and modulation types 4)

Advantages/disadvantages.

Keywords: modulation, demodulation, analog, digital, carrier wave, telephone line

1.13 Cable Modems

Definition: A cable modem gives Internet access through the cable TV (CATV) network, using coaxial cable and fibre (HFC - Hybrid Fibre-Coaxial).

Simple Explanation: The same cable that brings TV to your home also carries Internet data. The cable modem connects your computer to that cable.

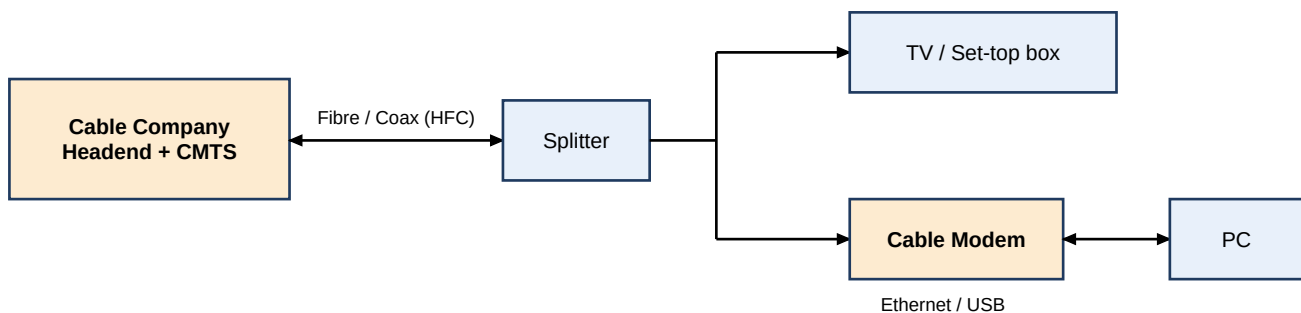


Fig 1.9: Cable modem connection

- **CMTS** (Cable Modem Termination System) is placed at the cable company headend; the cable modem is at the user end.
- Cable bandwidth is divided: **downstream video** 54-550 MHz, **downstream data** 550-750 MHz, **upstream data** 5-42 MHz.
- Standard used: **DOCSIS** (Data Over Cable Service Interface Specification). Textbook speeds: about 30 Mbps downstream and 12 Mbps upstream per channel; modern versions are much faster.
- Bandwidth is **shared** by all users on the same cable segment.

Point	Telephone (dial-up) modem	Cable modem
Medium	Telephone line (twisted pair)	Cable TV line (coaxial/HFC)
Speed	Up to 56 kbps	Mbps range
Connection	Dial-up; phone line busy	Always on; TV and Internet together
Bandwidth	Dedicated to the user	Shared among neighbours

Advantages: High speed, always connected, no phone line needed.

Disadvantages: Speed falls when many users are online; needs cable TV service; security risk due to sharing.

UNIT I QUICK REVISION

Topic	Remember
Definitions	Data communication = exchange of data via a medium. Distributed processing = task shared by many computers. Topology = arrangement of nodes and links. Protocol = rules of communication. Modem = modulator + demodulator.
Concepts	5 components: message, sender, receiver, medium, protocol. Line config: point-to-point, multipoint. Modes: simplex, half, full duplex. Networks: LAN, MAN, WAN.
Formula	Mesh links = $n(n-1)/2$; ports per device = $n-1$.
Differences	LAN/MAN/WAN; OSI vs TCP/IP (7 vs 4 layers); telephone modem vs cable modem; simplex vs half vs full duplex.
Diagrams	Data communication components, line configuration, topologies, OSI layers, OSI vs TCP/IP mapping, DTE-DCE, modem, cable modem.
OSI order	Physical, Data Link, Network, Transport, Session, Presentation, Application. PDUs: bits, frame, packet, segment, data.
Keywords	interoperability, ISO, IEEE, ITU-T, RS-232, DTE, DCE, ASK/FSK/PSK/QAM, CMTS, HFC, DOCSIS

UNIT I: POSSIBLE PRACTICE QUESTIONS

1. **(2 marks)** Define data communication. Name its five components.
2. **(2 marks)** What is distributed processing? Write two advantages.
3. **(Short)** Differentiate point-to-point and multipoint line configuration with diagrams.
4. **(Long)** Explain any four network topologies with diagrams, advantages and disadvantages.
5. **(Numerical)** A mesh network has 8 devices. Find the number of links and ports per device. (Ans: 28 links, 7 ports)
6. **(Short)** Compare LAN, MAN and WAN. Also explain simplex, half-duplex and full-duplex with examples.
7. **(Long)** Explain the OSI model with a diagram and describe the function of each layer.
8. **(Long)** Compare the OSI and TCP/IP models. Explain the layers of TCP/IP.
9. **(Short)** Write short notes on DTE-DCE interface and RS-232.
10. **(Short)** What is a modem? How is a cable modem different from a telephone modem?

UNIT II: Transmission Media and Channel Performance

2.1 Transmission Media: Introduction

Definition: Transmission media are the physical paths between sender and receiver through which data travels as signals.

Simple Explanation: Media can be **guided** (signal follows a solid path such as a cable) or **unguided** (signal spreads in air/space, no cable).

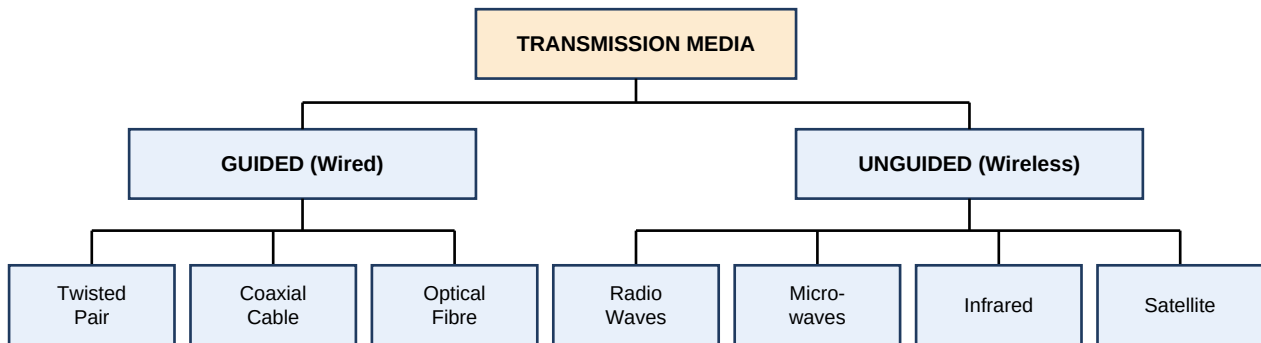


Fig 2.1: Classification of transmission media

2.2 Guided Media

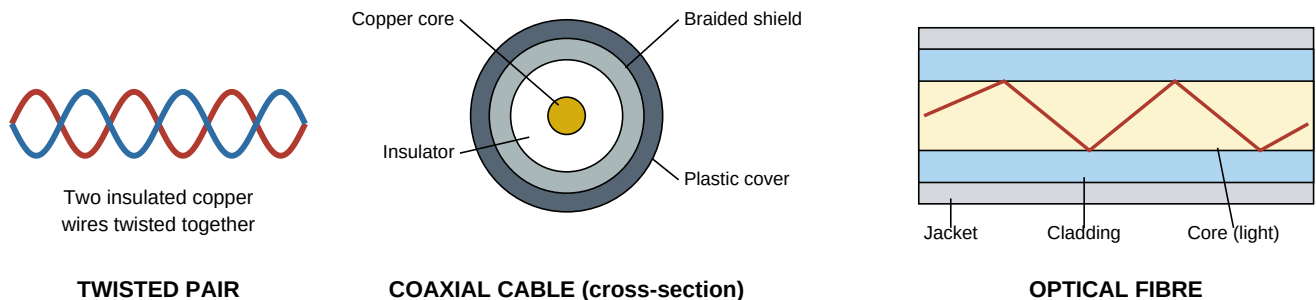


Fig 2.2: Structure of twisted pair, coaxial cable and optical fibre

(a) Twisted Pair Cable

Definition: Two insulated copper wires twisted together. Twisting cancels most noise and crosstalk from outside.

Point	UTP (Unshielded)	STP (Shielded)
Shield	No metal shield	Metal foil/braid around the pairs
Noise protection	Lower	Better
Cost / size	Cheap, thin, easy to install	Costlier, heavier
Use	Telephone lines, Ethernet LAN (RJ-45 connector)	Places with more interference

Categories of UTP: Cat 3 (10 Mbps), Cat 5 (100 Mbps), Cat 5e/Cat 6 (1 Gbps). Usual length limit is 100 m per segment.

Advantages: Cheap, flexible, easy to install.

Disadvantages: Affected by noise (UTP), short distance, limited bandwidth.

Applications: Telephone network, LAN, DSL.

(b) Coaxial Cable

Definition: A cable with a central copper core, insulator, outer braided conductor (shield) and plastic cover.

- Better noise immunity and higher bandwidth than twisted pair.
- Types: **RG-59** (75 ohm, cable TV), **RG-58** (50 ohm, thin Ethernet), **RG-11** (50 ohm, thick Ethernet). Connector: BNC.

Advantages: Good shielding, supports higher frequencies.

Disadvantages: Bulky, costlier than twisted pair, attenuation at high frequency needs repeaters.

Applications: Cable TV, cable modems, thin/thick Ethernet, telephone trunk lines.

(c) Optical Fibre

Definition: A thin glass or plastic cable that carries data as pulses of light.

Working: Light travels in the **core** by **total internal reflection**: the core has a higher refractive index than the **cladding**, so light hitting the boundary at more than the critical angle is fully reflected back. A **jacket** protects the fibre.

Source: LED or laser; detector: photodiode.

Mode	Description
Multimode step-index	Many light rays; sharp change of index between core and cladding; more distortion
Multimode graded-index	Refractive index changes gradually; less distortion than step-index
Single-mode	Very thin core, one ray, almost no distortion; longest distance and highest speed

Advantages: Very high bandwidth, low attenuation, immune to electromagnetic interference, light weight, secure, no corrosion.

Disadvantages: Costly; installation and repair are difficult; one-way (two fibres needed for two-way); fragile.

Applications: Backbone networks, long-distance telephone, HFC cable TV, high-speed LAN.

How to Write in Exam: 1) Definition 2) Diagram 3) Working 4) Advantages/disadvantages 5) Applications. For fibre, always write "total internal reflection".

Keywords: core, cladding, jacket, refractive index, critical angle, EMI, RJ-45, BNC, shield

2.3 Unguided Media (Wireless)

Definition: Unguided media carry signals as electromagnetic waves through air or space without any physical conductor.

Type	Frequency	Features	Uses
Radio waves	3 kHz - 1 GHz	Omnidirectional (all directions); pass through walls; can travel far	AM/FM radio, TV, cordless phone, paging
Microwaves	1 - 300 GHz	Unidirectional, line-of-sight (towers must face each other); affected by rain; parabolic dish antenna	Cellular phones, wireless LAN, telephone links
Infrared	300 GHz - 400 THz	Short range; cannot pass through walls, so no interference between rooms	TV remote, wireless keyboard/mouse
Satellite	Microwave bands	Satellite in orbit acts as relay (uplink and downlink); geostationary orbit is about 36,000 km high; delay is noticeable	TV broadcast, GPS, long-distance calls

Propagation modes: Ground (low frequency, follows earth), sky (reflected by ionosphere, up to about 30 MHz) and line-of-sight (above 30 MHz).

2.4 Transmission Impairments: Attenuation, Distortion, Noise

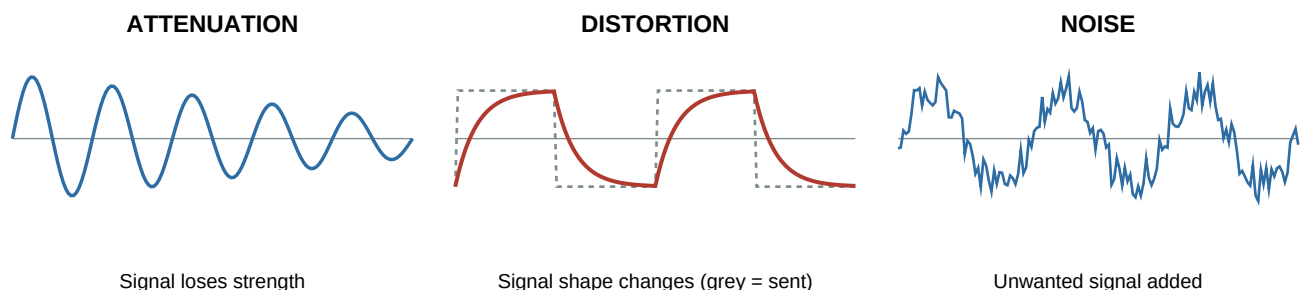


Fig 2.3: Attenuation, distortion and noise

Attenuation

Definition: Attenuation is the loss of signal strength (energy) as the signal travels through a medium. Energy is lost as heat due to resistance.

Remedy: Amplifiers or repeaters.

Formula: Attenuation (dB) = $10 \log_{10}(P_2 / P_1)$, where P_1 = input power, P_2 = output power. (For voltage use $20 \log_{10}(V_2/V_1)$.) A negative dB value means loss; a positive value means gain.

Solved example: A signal of 10 mW is reduced to 5 mW. dB = $10 \log_{10}(5/10) = 10 \log_{10}(0.5) = 10 \times (-0.301) = -3$ dB.

Common mistakes: Writing P_1/P_2 (wrong sign); using 20 instead of 10 for power; using natural log.

Distortion

Definition: Distortion is the change in the shape of a signal. It happens in composite signals because different frequency components travel at different speeds and reach the receiver at different times (delay distortion).

Remedy: Equalizers.

Noise

Definition: Noise is an unwanted signal added to the transmitted signal.

Type of noise	Cause
Thermal noise	Random motion of electrons in a wire
Induced noise	Motors and appliances acting like small antennas
Crosstalk	One wire affecting another wire nearby
Impulse noise	Sudden spike from power lines or lightning

Signal-to-Noise Ratio: SNR = average signal power / average noise power. $SNR_{dB} = 10 \log_{10}(SNR)$. Higher SNR means a better signal.

Solved example: Signal = 10 mW, noise = 0.001 mW. $SNR = 10 / 0.001 = 10,000$. $SNR_{dB} = 10 \log_{10}(10^4) = 40$ dB.

Common mistake: using different units for signal and noise.

Point	Attenuation	Distortion	Noise
Meaning	Loss of strength	Change of shape	Extra unwanted signal
Cause	Resistance of medium, distance	Different speeds of frequency components	Thermal, induced, crosstalk, impulse
Remedy	Amplifier/repeater	Equalizer	Shielding, twisting, filters

How to Write in Exam: 1) Definition 2) Cause 3) Effect/diagram 4) Formula and remedy. For noise, list its four types.

Keywords: loss of energy, decibel, delay distortion, thermal noise, crosstalk, SNR, amplifier, repeater

2.5 Throughput

Definition: Throughput is the actual number of bits that pass through a point in the network per second. Bandwidth is the maximum capacity; throughput is what you really get and is always equal or less.

Formula: Throughput = total bits transferred / time taken.

Solved example: A network sends 12,000 frames per minute; each frame has 10,000 bits. Throughput = $(12,000 \times 10,000) / 60 = 2,000,000$ bps = **2 Mbps**.

Common mistake: Forgetting to convert minutes to seconds.

2.6 Propagation Speed and Propagation Time

Definition: **Propagation speed** is the speed at which a signal travels through a medium (3×10^8 m/s in vacuum, lower in cable, about 2×10^8 m/s). **Propagation time** is the time taken by a signal to travel from one end of the link to the other.

Formula: Propagation time = Distance / Propagation speed. Also, Transmission time = Message size / Bandwidth.

Solved example: Distance = 12,000 km = 12×10^6 m, speed = 2.4×10^8 m/s. Propagation time = $12 \times 10^6 / 2.4 \times 10^8 = 0.05$ s = **50 ms**.

For a 2,500-byte message on a 1 Gbps link: transmission time = $(2,500 \times 8) / 10^9 = 0.00002$ s = 0.02 ms. Here propagation time is much larger, so distance controls the delay.

Common mistakes: Not converting km to metres; mixing bytes and bits (1 byte = 8 bits).

2.7 Wavelength

Definition: Wavelength is the distance a signal travels in one period (one complete cycle).

Formula: Wavelength = Propagation speed x Period = Propagation speed / Frequency (period = 1 / frequency).

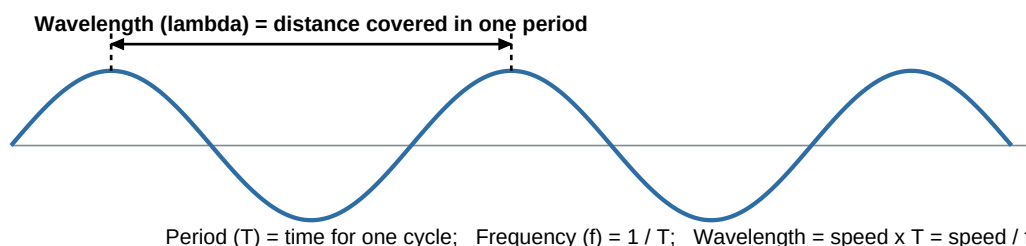


Fig 2.4: Wavelength of a signal

Solved example: An FM radio wave has frequency 100 MHz. Speed in air = 3×10^8 m/s. Wavelength = $3 \times 10^8 / (100 \times 10^6) = 3$ m.

Common mistake: Using MHz directly; convert to Hz first. Higher frequency means shorter wavelength.

2.8 Shannon Capacity

Definition: Shannon capacity is the theoretical maximum data rate (bits per second) of a noisy channel. It was given by Claude Shannon.

Formula: $C = B \times \log_2(1 + \text{SNR})$, where C = capacity (bps), B = bandwidth (Hz), SNR = signal-to-noise ratio as a plain ratio (not in dB).

- Capacity depends only on bandwidth and SNR, not on the number of signal levels.
- It is an upper limit; real channels give less. If SNR = 0, capacity = 0.
- To convert dB to ratio: $\text{SNR} = 10^{(\text{SNR}_{\text{dB}}/10)}$. Tip: $\log_2(x) = 3.32 \times \log_{10}(x)$.

Solved example 1: B = 1 MHz = 10^6 Hz, SNR = 63. $C = 10^6 \times \log_2(1 + 63) = 10^6 \times \log_2(64) = 10^6 \times 6 = 6$ Mbps.

Solved example 2 (SNR in dB): Telephone line: B = 3000 Hz, $\text{SNR}_{\text{dB}} = 35$. $\text{SNR} = 10^{3.5} = 3162$. $C = 3000 \times \log_2(3163) = 3000 \times 11.63 = \text{about } 34.9$ kbps.

Common mistakes: Using SNR_{dB} directly in the formula; forgetting the "1 +"; using $\log_{10}$ instead of $\log_2$.

How to Write in Exam: 1) Definition 2) Formula with meaning of each symbol 3) Solved numerical 4) Final answer with unit.

Keywords: noisy channel, theoretical maximum, bandwidth, SNR, $\log_2(1+\text{SNR})$, bps

2.9 Comparison of Transmission Media

Medium	Data rate	Cost	Noise / EMI	Distance	Typical use
UTP	Low to medium (up to about 1 Gbps)	Lowest	Affected	About 100 m	Telephone, LAN
STP	Similar to UTP	Medium	Better than UTP	About 100 m	LAN in noisy area
Coaxial	Medium	Medium	Good	Hundreds of m	Cable TV, old Ethernet
Optical fibre	Very high	High	Immune to EMI	Many km	Backbone, long distance
Radio waves	Low to medium	Low	Interference possible	Long range	Broadcast, cordless phone
Microwaves	High	Medium	Weather (rain) affects	Line-of-sight, towers	Cellular, telephone links
Infrared	Low	Low	Blocked by walls	A few metres	Remote controls
Satellite	High	Very high	Weather affects	Global	TV, GPS

Guided vs Unguided: Guided: signal follows a physical path, more secure, limited by cable; Unguided: signal spreads in air, easy to install and mobile, but less secure and more affected by weather and interference.

UNIT II QUICK REVISION

Topic	Remember
Definitions	Attenuation = loss of strength. Distortion = change of shape. Noise = unwanted signal. Throughput = actual bits per second. Wavelength = distance travelled in one period. Shannon capacity = maximum data rate of a noisy channel.
Concepts	Guided: twisted pair, coaxial, fibre. Unguided: radio, microwave, infrared, satellite. Fibre works by total internal reflection. Noise types: thermal, induced, crosstalk, impulse.
Formulas	$dB = 10 \log_{10}(P_2/P_1)$. $SNR_{dB} = 10 \log_{10}(SNR)$. Throughput = bits / time. Propagation time = distance / speed. Transmission time = size / bandwidth. Wavelength = speed / frequency. $C = B \log_2(1 + SNR)$.
Differences	UTP vs STP; guided vs unguided; attenuation vs distortion vs noise; bandwidth vs throughput.
Diagrams	Media classification, twisted pair/coax/fibre structure, attenuation-distortion-noise, wavelength.
Keywords	guided, unguided, EMI, decibel, repeater, equalizer, SNR, latency, line-of-sight, geostationary, core, cladding

UNIT II: POSSIBLE PRACTICE QUESTIONS

1. **(2 marks)** Define transmission media. Classify them with a diagram.
2. **(Short)** Differentiate UTP and STP cables.
3. **(Long)** Explain the structure and working of optical fibre. Write its advantages and disadvantages.
4. **(Short)** Write short notes on radio waves, microwaves and infrared.
5. **(Numerical)** A signal power falls from 10 mW to 2.5 mW. Find the attenuation in dB. (Ans: -6.02 dB)
6. **(Short)** Differentiate attenuation, distortion and noise. Name the types of noise.
7. **(Numerical)** Find the propagation time for a 3000 km link with speed 2×10^8 m/s. (Ans: 15 ms)
8. **(Numerical)** Find the wavelength of a 50 MHz signal in vacuum. (Ans: 6 m)
9. **(Numerical)** State Shannon capacity formula. Find capacity for $B = 4$ kHz and $SNR = 255$. (Ans: $4000 \times \log_2 256 = 32$ kbps)
10. **(Long)** Compare all transmission media in a table (data rate, cost, noise immunity, distance, use).